

continuously.

The next thing you need to know is the measurement system. In the context of process control, controller decisions are based on measurements of process parameters. With the advent of computers, it is now possible to do inferential measurements, which means telling the value of a parameter without actually measuring it physically. It should, however, be remembered that inferential measurement algorithms are also based on physical measurements. Therefore, rather than rendering measurements redundant, they have made measurements all the more important.

Final control elements can refer to three things: control valves, variable-speed drives and dampers. In process plants, more often than not, the final control element is the control valve.

manufacturing automation applications to facilitate interoperability.

Try to realise the practical utilities of the standardisations—the user's project cycle is shorter using standardised software components and their cost is lower. These benefits are real and tangible. Because the OPC standards are based, in turn, upon computer industry standards, technical reliability is assured.

The original specification standardised the acquisition of process data. It was quickly realised that communicating other types of data could benefit from standardisation. So standards for alarms and events, historical data and batch data were launched.

Additionally, you need to know the current and emerging OPC specifications and their applicability. The original OPC Data Access used to move

vendor interoperability. It also adds remote configuration, diagnostic and monitoring/management services.

OPC Historical Data Access provides access to data already stored. From a simple serial data logging system to a complex SCADA system, historical archives can be retrieved in a uniform manner.

Moreover, all standardisations ensure secured mode of operation—OPC Security specifies how to control client access to servers in order to protect this sensitive information and to guard against unauthorised modification of process parameters. All these commands allow the users to identify, send and monitor control commands which execute on a device.

## How to communicate for better control?

Although computers, PLCs and remote terminal units communicate with each other digitally, most end devices (valves, pressure transducers, switches, etc) still use analogue signals. For example, an analogue value of 4 mA might correspond to a pressure of no flow, while a value of 20 mA might correspond to a 1000GPM flow value. With discrete devices, the presence of a signal might represent a 'closed' or 'alarm' condition, while the absence of a signal might represent 'open' or 'normal.'

But keep in mind, in the future, the 4-20mA standard will be replaced with a digital, two-way, multidrop communication—FieldBus. You need to know the reason behind that. In two-way communications, a value can not only be read from the end device but also be written to the device. For example, the calibration constants associated with a particular sensor can now be stored directly in the device itself and changed as needed. The multi-drop capability of a FieldBus results in the most immediate cost savings for users.

With analogue devices, a separate cable needs to be run between the end device and the control system because only a single analogue signal can be

## The automation industry is growing 6-8 per cent globally, whereas in countries like India and China, it is growing at more than 30 per cent

The issues relating to final control elements are most relevant to control valves, although these are applicable to a large extent to dampers and in some cases variable-speed drives as well. So try to get a clear idea about the instrumentation part of the control valve.

## How to standardise the control?

For starters, learn to use OPC—'OLE (object linking and embedding technology) for process control.' It is a series of standards' specifications. The first standard—originally called the OPC specification and now the Data Access specification—resulted from the collaboration of a number of leading worldwide automation suppliers working in cooperation with Microsoft. Originally based on Microsoft's OLE COM (component object model) and DCOM (distributed component object model) technologies, the specification defined a standard set of objects, interfaces and methods for use in process control and

real-time data from PLCs, distributed control systems and other control devices to human machine interface and other display clients. The Data Access 3 specification is now a Release Candidate. It leverages earlier versions while improving the browsing capabilities and incorporating XML-DA Schema.

OPC Alarms & Events provides alarm and event notifications on demand (in contrast to the continuous data flow of Data Access). These include process alarms, operator actions, informational messages and tracking/auditing messages.

OPC Batch specification carries the OPC philosophy to the specialised needs of batch processes. It provides interfaces for the exchange of equipment capabilities (corresponding to the S88.01 physical model) and current operating conditions.

Client-to-server and server-to-server communication across Ethernet fieldbus networks is facilitated by OPC Data exchange. It provides multi-